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An overview of the capabilities of ChatGPT for medical writing and its implications for academic integrity

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Abstract

The artificial intelligence (AI) tool ChatGPT, which is based on a large language model (LLM), is gaining popularity in academic institutions, notably in the medical field. This article provides a brief overview of the capabilities of ChatGPT for medical writing and its implications for academic integrity. It provides a list of AI generative tools, common use of AI generative tools for medical writing, and provides a list of AI generative text detection tools. It also provides recommendations for policymakers, information professionals, and medical faculty for the constructive use of AI generative tools and related technology. It also highlights the role of health sciences librarians and educators in protecting students from generating text through ChatGPT in their academic work.

KEYWORDS

artificial intelligence (AI); librarians, health science; libraries, academic; plagiarism; students, medical

INTRODUCTION

The history of artificial intelligence (AI) dates back to the 1950s when researchers started investigating the idea of AI. ELIZA, the first AI program, which was created in the beginning stages of AI development to simulate human conversation. The development of more complex chatbots that could comprehend and respond to complex requests was made possible by the development of AI technology over the subsequent decades. As technology advances, AI and chatbots started being used more and more in a variety of

industries, including customer services, education, and healthcare (King, 2023a).

AI chatbots have been used to build interfaces that can converse with users, respond to questions, and comprehend and act upon natural language inputs. Questions about the potential applications and effects of AI in language processing have been raised in light of its capacity to produce credible content (Cotton et al., 2023). An AI-powered tool called ChatGPT was created by OpenAI and released in November 2022. Within the first 2 months of its launch ChatGPT had more than 30 million users and received about 5 million daily visits. This is one of the fastest ever growing software products in history (Sun & Hoelscher, 2023). It provides a straightforward textual interface as a natural language processing and question-answering system that allows users to quickly create simple and direct prompts, ask questions, and request the generation of different text

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types. It has a large and complex dataset that gives it the ability to learn and predict the next word in a sentence in an incredible way. ChatGPT retrieves information from various Internet sources. It covers a wide range of subjects and can generate different information types such as essays, outlines, abstracts, reports etc. (Shiri, 2023). Shah (2022) reported that there has been a weekly increase in AI tools, and that experts have been deceived by AI tools as they have become so competent in answering queries.

AI generative tools can expand texts to become more detailed and in-depth; format and structure essays and reports, provide information summaries and run grammar and spell-checks. They also provide the facility of rewriting the same sentence repeatedly. They can be used to find ideas, create plots, and generate outlines; see Table 1.

Medical students' use of AI/ChatGPT

Medical students have made use of AI generative tools for writing assignments, essays, reports, scientific papers, case reports and seeking medical information; see Table 2. The accuracy of the content is still in question, and frequently citations generated appear incorrect. There are a number of other risks associated including transparency or ethical issues, plagiarism, risk of bias, inadequate depth of information, lack of human insight, risk of spreading

misinformation, and incomplete understanding of medical concepts (Gordijn & Have, 2023; Mbakwe et al., 2023; O'Connor, 2022).

AI tools will play a significant role in medicine (Park et al., 2020); however, it is unclear how this will impact medical students. Mehta et al. (2021) who measured perception of medical students about the role of AI, reports that the most students think that AI will take over jobs in terms of providing medical documentation, planning, and resource management, but did not see that AI could replace doctors. Sedaghat (2023) reported that the performance of ChatGPT to pass a medical exam is at an acceptable level and it has capability to change medical practices. A study by Friederichs et al. (2023) reported that ChatGPT outperformed almost all medical students in first, second, and third years in the German State exam in Progress Test Medicine. Only students from fourth years and beyond were able to achieve comparable, but not higher scores.

A lack of education on AI can result in a lack of understanding towards AI in medicine (Pinto dos Santos et al., 2019). Doumat et al. (2022) measured medical students' knowledge, attitudes and perception towards AI. Researchers also looked at how AI could be incorporated into the medical curriculum as a teaching tool. Data collected from medical institutes indicated that (81.1%) of respondents wanted AI incorporated into the curriculum. Students who were in their clinical years are more knowledgeable about AI in medicine. When asked about

TABLE 1 Example AI generative tools.

Wordtune http://www.wordtune.com	Rewrites text and suggests improvements.	Subscription based. A free option with limited functionality is available.
Google Bard AI bard.google.com	Bard is Google's AI chatbot in the style of ChatGPT. It helps writing text including assignments, articles and blogs.	Currently free to use.
Jasper https://www.jasper.ai/	AI writing tool.	Subscription based. There is a free 7-day trial.
copy.ai https://www.copy.ai/	AI content generator.	Subscription based. A free option allows up to 2000 words per month.
Writesonic https://www.writesonic.com/	AI-based writer, supported by search engine optimization (SEO).	Subscription based. A free trial offers 10,000 words per month.
Ryter https://rytr.me/	AI writing assistant. Suggests rephrases.	Subscription based. The free plan offers 10,000 characters per month.
Writer https://writer.com/	Generates article outlines and contents, offers feedback on texts.	Subscription based
Anyword https://anyword.com/	Copywriting tool for creating blogs, websites, articles.	Subscription based. Offers a 7-day free trial.
Sudowrite https://www.sudowrite.com	AI writing tool. More suitable for beginners.	Subscription based.

TABLE 2 ChatGPT/AI use among medical students.

ChatGPT/AI use	Description
Assignment Writing	Students may write information for assignments without understanding the context. Thus, both learning process and academic integrity are compromised.
Essays and Reports Writing	The use of AI tools to write essays, reports, or other pieces of writing is considered a form of plagiarism.
Personalized Learning	Students can ask questions about topics in medicine and receive clear, brief responses to develop their initial understanding about the topic. AI has the potential to help students understand medical terminologies, and help them in their personalized learning.
Medical Information	Using AI to seek medical information to create literature reviews raises concerns about plagiarism, lack of originality, inaccurate content, bias, and understanding of medical concepts.
Case Report/Patient Clinic Letters	ChatGPT has been used for preparing case, or writing patients clinic letters.

their preferred methods of learning suturing, 27.9% of the participants selected surgeons as teachers, only 1.5% selected artificial intelligence, and 70.6% selected a mix of AI and surgeons. The study recommended that AI education should be included in the curriculum of medical schools. Adoption of AI as a tool in medical education will improve medical students' knowledge of AI, as well as unlocking AI's potential to transform medical education.

ChatGPT and healthcare practice

A recent review showed that ChatGPT has potential to help streamline clinical workflows, potentially saving cost and creating efficiencies in healthcare delivery (Gunawan, 2023; Khan et al., 2023; Shen et al., 2023). ChatGPT can create discharge summaries, reducing the burden of documentation in healthcare (Patel & Lam, 2023), and has the potential to revolutionize healthcare by improving diagnostics, predicting disease risk and outcomes, and discovering new drugs (Mann, 2023; Sharma & Thakur, 2023). ChatGPT can also improve health literacy by making health information easily accessible to the public.

Medical professionals have a responsibility to provide patients with evidence-based care, but it becomes challenging to keep up with the pace of information generation. Ultimately, this can affect the quality of patient care (Azam & Naeem, 2022; Friederichs et al., 2023). AI tools could potentially help with this. Despite opportunities with ChatGPT, there are concerns with its use such as transparency issues (Holzinger et al., 2023), inaccurate content leading to incorrect decisions (Rao et al., 2023), and legal and accountability issues in case of medical errors caused through ChatGPT (Biswas, 2023).

ChatGPT and health education

Due to the personalized interaction that ChatGPT provides, it can enable powerful self-learning (Fijacko et al., 2023). It can serve as a teaching assistant with its capacity to produce exercises, tests, and scenarios that can be used in the classroom. Its capacity to produce translations and summaries can also be utilized to make learning materials more comprehensible for students (Anders, 2023; Gao et al., 2023). Furthermore, it can assist faculty with access to information, documentation, decision support etc. (Jeblick et al., 2022; King, 2023b). However, there are several concerns about the quality of outputs, such as biased or inaccurate information, and that information is limited to the period before the year 2021. It is also currently unable to handle images (Gilson et al., 2023).

ChatGPT and academic integrity

The use of ChatGPT raises concerns regarding academic integrity if students use it for assignments. Students may also become overly dependent on it, which could result in a decline in original thought, self-directed learning, critical thinking abilities, and writing skills (Sun & Hoelscher, 2023).

Academic integrity ensures quality, authenticity and validity in the field of medicine, where students are trained to become honest medical professionals. No one would like to be treated by a doctor who had cheated in an exam (Błachnio et al., 2021). A survey conducted by Intelligent (2023) revealed that university students use ChatGPT for academic writing, and although 75% considered it a form of cheating, they continue to use it. Donmez et al. (2023) experimented with ChatGPT and concluded that as ChatGPT diminishes the distinction between humans and machine writing, it would be difficult to analyse whether research was written by a human or an AI tool. Moreover, Dwivedi et al. (2023) reported that some articles are easy to write for ChatGPT, and that

TABLE 3 AI generated text detection tools.

AI text detectors	Description
Turnitin AI Detection https://www.turnitin.com/solutions/ai-writing	Turnitin introduced AI detection to help educators identify when AI generated content in students' assignment submissions. Subscription based.
Originality.ai https://originality.ai/	AI content detector and plagiarism checker. It has the most accurate ChatGPT, Bard, paraphrasing, and GPT-4 AI detector built. Subscription based.
Copyleaks https://copyleaks.com/	AI-based text analysis tool to detect AI content and help prevent plagiarism.
AI Detector Pro https://aidetector.pro/	Indicates content which is clearly or possibly AI generated. Subscription based
GPTZero https://gptzero.me/	Detects text generated through ChatGPT, BARD, and other AI models. Subscription based.
ZeroGPT https://www.zerogpt.com/	AI detector, which detects OpenAI text. Subscription based.
Crossplag https://crossplag.com/ai-content-detector/	AI content detector. Aims to detect the originality of the text.
OpenAI https://openai.com/blog/new-ai-classifier-for-indicating-ai-written-text	OpenAI indicates AI-written text. Subscription based.
Sapling https://sapling.ai/ai-content-detector	AI detector. Subscription based.
KAZAN SEO https://kazanseos.com/	KAZAN SEO helps discover AI generated text.
Writer https://writer.com/ai-content-detector/	Can check up to 1500 characters. Currently only available in the writer app as an API. Free.

in near future ChatGPT would be able to analyse data, and produce theories more efficiently than a human. To ensure academic integrity, the New York Education Department completely banned the usage of AI tools on network devices (Yang, 2023), and an international conference on machine learning (ICML) has banned authors and their scientific writing created through AI generative tools (Vincent, 2023).

Tools to detect AI generative text

Several AI tools have been developed that can be used by students, research advisors, teachers, journals editors, and others to detect the ChatGPT generated texts; see Table 3. Most of these tools are subscription based.

DISCUSSION

There is concern among faculty members on the challenge of academic integrity in the era of ChatGPT (Cotton et al., 2023). AI tools create threats for both students and teachers. Creating risk of biased, misleading information, and plagiarism (O'Connor, 2022); the spread of misinformation (Lubowitz, 2023); violation of copyright (Gordijn & Have, 2023); and academic misconduct (Health, 2023). It also creates challenges in handling complex scenarios in academia, because AI models generate content using algorithms and previous information rather than critical thinking, human insight and thoughtful reasoning (Mbakwe et al., 2023). A debate has erupted regarding whether AI generative tools should be banned for academic writing, though some argue they present an opportunity to reconsider current teaching methods. Rather than simply ignoring the capabilities and potential of AI applications, educators could assign different tasks to students. One example is to ask students to critically evaluate AI generated content (Baidoo-Anu & Ansah, 2023; García-Peñalvo, 2023; Haensch et al., 2023). Another strategy may be to design tests that require students to demonstrate communication skills, problem-solving, and critical-thinking abilities (Shiri, 2023), or to design tests that promote creativity and originality, or that ask for image interpretation, or responses to clinical scenarios. Medical educators could create exams that require students to formulate their own research questions, or to create and support their own arguments. Designing assignments that demands a comprehensive contextual information, human insight, and self-learning will decrease students' dependency on ChatGPT (Sun & Hoelscher, 2023). It is crucial to use AI in educational contexts in an ethical and responsible manner after considering the potential limitations and the significance of encouraging students in critical thinking.

Every person has different learning requirements, aptitudes, and preferences, and ChatGPT can offer opportunities for learning at all levels. It has the potential to enhance students' reading and critical thinking and makes reading easier by simplifying complex text. It can provide solutions to queries and can help with conceptualization. It can improve students' research skills

by presenting summaries of large texts. It can help in the development of a variety of professional skills, such as project management and decision-making (Kasneci et al., 2023).

According to AlAfnan et al. (2023), ChatGPT has the potential to replace search engines and can transform learning from theory-based to application-based ideas and contents. Recent studies indicate that the users are more likely to use AI-powered tools over search engines in the future for searching information on the Internet. Although, currently ChatGPT is unable to crawl the Internet, it can summarize the content. AI tools can be used to summarize the contents of selected or top search results (Noorden, 2023; Zhao et al., 2023). If AI tools obtain the capability of crawling the internet in the future they will be more popular than traditional internet search tools.

Implications for policy makers, health sciences librarians

Health sciences librarians have a long history of providing information literacy instruction. The importance of AI literacy in medical teaching and learning cannot be understated in the technologically fast-growing changing environment. AI literacy is defined as 'a multifaceted concept that involves a comprehensive understanding of AI, its applications in education, its ethical implications, and the ability to engage in transparent and authentic dialogue about AI' (Cardona et al., 2023). There is a need to develop the AI literacy of medical students. Health sciences librarians should organize seminars and awareness sessions to demonstrate AI's beneficial and constructive uses and make clear the risks involved. Health sciences librarians should continue to educate students about the responsible use of information resources and validity of scientific evidence through the use of trustworthy medical databases, journals and eBooks. Librarians could disseminate the AI generated text detection tools to help reduce the use of ChatGPT for generating texts for assignments, and research related tasks among students. There is also a clear need for health science librarians' and educators' involvement in AI implementation and usage guidelines. The involvement of health sciences librarians and educators is important to allow AI guidance and processes to be aligned for a shared vision for education and research. Moreover, the involvement of health sciences librarians will increase trust on the information retrieved through AI generative tools. AI use guidelines can be developed by collaborating with all stakeholders such as, educators,

course instructors, and research advisors. The guidelines will support the constructive use of AI generative tools among students, describing dos and don'ts for its use within assignments.

CONCLUSION

ChatGPT has a potential to transform medical students' capabilities in writing, and professional development, when used responsibly. There are risks associated with using ChatGPT, but these can be minimized by effectively implementing AI use guidelines, developing AI literacy, sharing knowledge, and applying AI generative text detection tools.

AI text generative tools raise concerns about potential misuse. The proactive adoption of large language models (LLMs) technologies, combined with careful consideration of potential ethical and legal issues, can help in limiting the future complications. ChatGPT has the potential to accelerate innovation in healthcare by overcoming language barriers. Therefore, a debate on the pros and cons of ChatGPT is strongly recommended, and its potential benefits should be balanced against the risks of misleading results, fraudulent research, and scholarly contents generated through AI.

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CONFLICT OF INTEREST STATEMENT

There is no conflict of interest.

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